Scientific Calculator in C

Sample Output

Program Startup

```
╔════════════════════════════════════════════╗
║         SCIENTIFIC CALCULATOR              ║
║   Type 'help' for operations               ║
╚════════════════════════════════════════════╝

>
```

Help Menu

```
> help

  Basic:   + - * / mod pow
  Trig:    sin cos tan asin acos atan (degrees)
  Log:     log log2 log10 exp
  Other:   sqrt abs ceil floor round
  Mem:     ms (store) mr (recall) mc (clear)
  Const:   pi e
  hist:    history  |  exit: quit
```

Addition

```
> +

a = 25
b = 15

= 40
```

Power Operation

```
> pow

a = 2
b = 8

= 256
```

Square Root

```
> sqrt

a = 144

= 12
```

Trigonometric Function

```
> sin

a = 30

= 0.5
```

```
> cos

a = 60

= 0.5
```

Logarithm

```
> log10

a = 1000
```

```
= 3
```

Constants

```
> pi

π = 3.1415926536
```

```
> e

e = 2.7182818285
```

Memory Operations

```
> pow

a = 5
b = 2

= 25
```

```
> ms

Stored: 25
```

```
> mr

Recalled: 25
```

```
> mc

Memory cleared
```

Calculation History

```
> hist

=== Calculation History ===

  [1] = 40
  [2] = 256
  [3] = 12
  [4] = 0.5
  [5] = 0.5
  [6] = 3
  [7] = 3.141592654
  [8] = 2.718281828
  [9] = 25
```

Error Handling

Division by Zero

```
> /

a = 50
b = 0

Error: Division by zero!
```

Square Root of Negative Number

```
> sqrt

a = -25

Error: sqrt of negative!
```

Exit

```
> exit

Bye!
```

Features Demonstrated

- Basic Arithmetic Operations
- Power and Modulus Calculations
- Trigonometric Functions
- Logarithmic and Exponential Functions
- Mathematical Constants (π and e)
- Memory Store, Recall and Clear
- Calculation History
- Error Handling
- Interactive Command-Based Interface

Author

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